

1. A truck of mass 1500 kg is moving on a straight horizontal road. The engine of the truck is working at a constant rate of 30 kW . The resistance to the motion of the truck is modelled as a constant force of magnitude R newtons. At the instant when the truck is moving at a speed of 20 m s^{-1} , the acceleration of the truck is 0.6 m s^{-2}

(a) Find the value of R .

(4)

Later on, the truck is moving up a straight road that is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{8}$

The resistance to the motion of the truck from non-gravitational forces is modelled as a constant force of magnitude 500 N .

The engine of the truck is again working at a constant rate of 30 kW .

At the instant when the speed of the truck is $V\text{ m s}^{-1}$, the deceleration of the truck is 0.2 m s^{-2}

(b) Find the value of V

(4)

2. A particle P of mass 0.5 kg is moving with velocity $(5\mathbf{i} + 3\mathbf{j})\text{ m s}^{-1}$.
The particle receives an impulse $(-2\mathbf{i} + \lambda\mathbf{j})\text{ N s}$, where λ is a constant.
Immediately after receiving the impulse, the velocity of P is $(x\mathbf{i} + y\mathbf{j})\text{ m s}^{-1}$.
The kinetic energy gained by P as a result of receiving the impulse is 22 J .
Find the possible values of λ .

(7)

3.

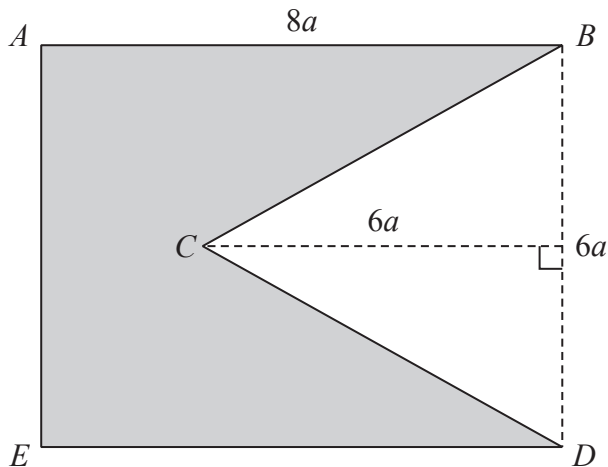


Figure 1

The uniform lamina $ABDE$ is in the shape of a rectangle with $AB = 8a$ and $BD = 6a$. The triangle BCD is isosceles and has base $6a$ and perpendicular height $6a$.

The template $ABCDE$, shown shaded in Figure 1, is formed by removing the triangular lamina BCD from the lamina $ABDE$.

(a) Show that the centre of mass of the template is $\frac{14}{5}a$ from AE .

(5)

The template is freely suspended from A and hangs in equilibrium with AB at an angle of θ° to the downward vertical.

(b) Find the value of θ , giving your answer to the nearest whole number.

(3)

4. [In this question, the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.]

A particle Q of mass 1.5 kg is moving on a smooth horizontal plane under the action of a single force $\mathbf{F}$ newtons. At time t seconds ($t \geq 0$), the position vector of Q , relative to a fixed point O , is $\mathbf{r}$ metres and the velocity of Q is $\mathbf{v}\text{ ms}^{-1}$

It is given that

$$\mathbf{v} = (3t^2 + 2t)\mathbf{i} + (t^3 + kt)\mathbf{j}$$

where k is a constant.

Given that when $t = 2$ particle Q is moving in the direction of the vector $\mathbf{i} + \mathbf{j}$

(a) show that $k = 4$

(2)

(b) find the magnitude of $\mathbf{F}$ when $t = 2$

(4)

Given that $\mathbf{r} = 3\mathbf{i} + 4\mathbf{j}$ when $t = 0$

(c) find $\mathbf{r}$ when $t = 2$

(4)

5.

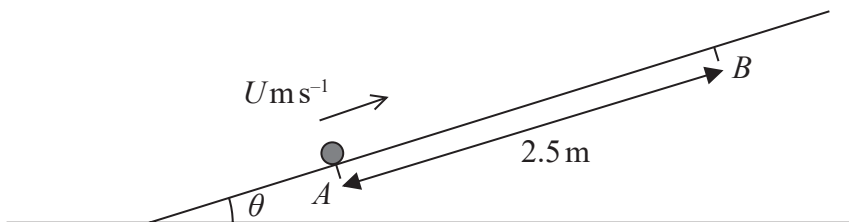


Figure 2

A rough straight ramp is fixed to horizontal ground. The ramp is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

The points A and B are on a line of greatest slope of the ramp, with $AB = 2.5 \text{ m}$ and B above A , as shown in Figure 2.

A package of mass 1.5 kg is projected up the ramp from A with speed $U \text{ m s}^{-1}$ and first comes to instantaneous rest at B .

The coefficient of friction between the package and the ramp is $\frac{2}{7}$

The package is modelled as a particle.

(a) Find the work done against friction as the package moves from A to B .

(3)

(b) Use the work–energy principle to find the value of U .

(4)

After coming to instantaneous rest at B , the package slides back down the slope.

(c) Use the work–energy principle to find the speed of the package at the instant it returns to A .

(3)

6.

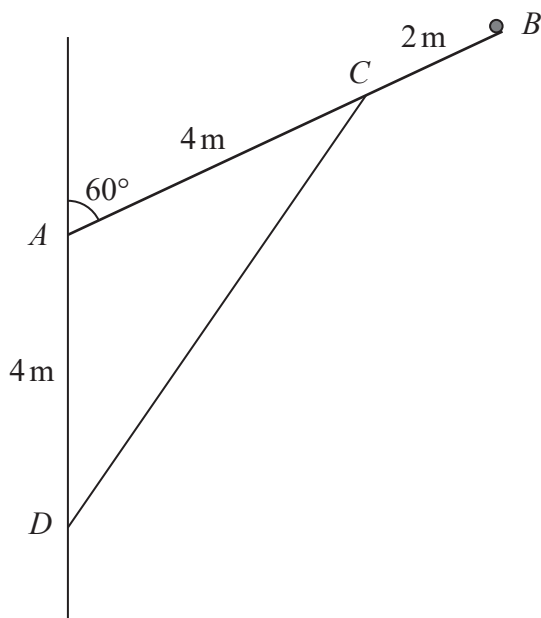


Figure 3

A uniform pole AB , of weight 50 N and length 6 m , has a particle of weight W newtons attached at its end B . The pole has its end A freely hinged to a vertical wall.

A light rod holds the particle and pole in equilibrium with the pole at 60° to the wall.

One end of the light rod is attached to the pole at C , where $AC = 4\text{ m}$.

The other end of the light rod is attached to the wall at the point D .

The point D is vertically below A with $AD = 4\text{ m}$, as shown in Figure 3.

The pole and the light rod lie in a vertical plane which is perpendicular to the wall.

The pole is modelled as a rod.

Given that the thrust in the light rod is $60\sqrt{3}\text{ N}$,

(a) show that $W = 15$

(4)

(b) find the magnitude of the resultant force acting on the pole at A .

(6)

7. Particle P has mass $3m$ and particle Q has mass km . The particles are moving towards each other on the same straight line on a smooth horizontal surface.

The particles collide directly.

Immediately **before** the collision, the speed of P is $2u$ and the speed of Q is $3u$.

Immediately **after** the collision, the speed of P is u and the speed of Q is v .

The direction of motion of P is unchanged by the collision.

(a) Show that $v = \frac{(3 - 3k)}{k} u$ (3)

(b) Find, in terms of m and u , the magnitude of the impulse received by Q in the collision. (2)

The coefficient of restitution between P and Q is e .

Given that $v \neq u$

(c) find the range of possible values of k . (5)

8. A particle P is projected from a fixed point O . The particle is projected with speed $u \text{ m s}^{-1}$ at angle α above the horizontal. The particle moves freely under gravity. At the instant when the horizontal distance of P from O is x metres, P is y metres vertically above the level of O .

(a) Show that $y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$

(6)

A small ball is projected from a fixed point A with speed $U \text{ m s}^{-1}$ at θ° above the horizontal.

The point B is on horizontal ground and is vertically below the point A , with $AB = 20 \text{ m}$.

The ball hits the ground at the point C , where $BC = 30 \text{ m}$, as shown in Figure 4.

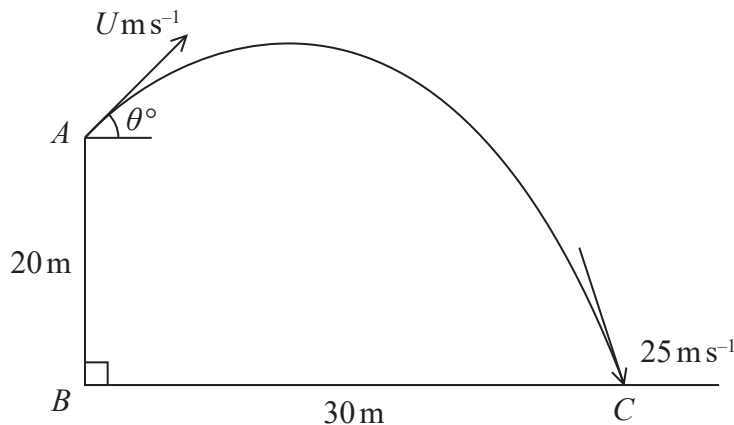


Figure 4

The speed of the ball immediately before it hits the ground is 25 m s^{-1}

The motion of the ball is modelled as that of a particle moving freely under gravity.

- (b) Use the principle of conservation of mechanical energy to find the value of U .

(3)

- (c) Find the value of θ

(3)